WORKFLOW AUTOMATION USING DELUGE

Student Admission Workflow Automation

Use Case Report

Prepared for academic submission

1. AIM

To implement workflow automation using Deluge in the Student Admission Workflow Automation system, reducing manual work and improving the accuracy and consistency of academic record handling, student profile management, document verification, interview scheduling, and admission decision processing.

2. OBJECTIVE

• Automate repetitive tasks in the student admission process using Deluge functions and workflow rules.

• Validate academic marks, passing marks, grade point averages, and required application fields.

• Maintain and update student application statuses based on workflow conditions.

• Support document verification by identifying valid, expired, failed, or not-submitted documents.

• Automate notifications or follow-up actions when documents are missing or applications require attention, where notification actions are configured.

• Coordinate interview slot details and admission decision records.

• Reduce data-entry errors and improve the speed, traceability, and reliability of admission processing.

3. WORKFLOW MAPPING

The following mapping describes a practical workflow for the modules visible in the supplied application screenshots. Exact trigger names and Deluge functions depend on the configuration of the Creator application.

Module / Form	Trigger or Event	Workflow Action
Student Profile	On form submission or edit	Validate required profile fields; set or preserve application status; flag incomplete submissions.
Academic Record	On form submission or edit	Check grade, total marks, passing marks, and GPA; validate certificate URL and verification status.
Document Upload	On upload or record update	Check document type and verification result; flag expired, invalid, or missing documents for review.
Interview Slots	On slot creation or update	Validate interview type, duration, date/time, and slot availability; update interview scheduling status.
Admission Decisions	On decision submission	Validate decision outcome, decision date, scholarship amount, enrollment deadline, and decision maker.
Dashboard / Reports	After records are updated	Reflect current totals, GPA averages, verification counts, interview distributions, and application-status charts.

4. ADDITIONAL WORKFLOW CONFIGURATION

• **Trigger selection:** Configure workflows to run when a record is created, edited, submitted, or when a scheduled follow-up is required.

• **Criteria:** Define clear conditions, such as verification result = Expired, application status = Submitted, GPA below a configured threshold, or an enrollment deadline approaching.

• **Field updates:** Update only the relevant status or review fields so existing data is not overwritten unintentionally.

• **Notifications:** Where supported and configured, notify the responsible admission staff member when documents need review, an interview is scheduled, or an enrollment deadline is near.

• **Scheduling:** Use scheduled workflows for reminders and periodic checks; confirm date/time zone and recurrence settings.

• **Permissions:** Ensure workflow actions run with appropriate access and users can view only authorized student information.

• **Error handling:** Handle blank values, invalid URLs, unexpected statuses, and missing related records; log failures where suitable.

• **Testing:** Test accepted, rejected, incomplete, expired-document, and boundary-value scenarios using sample data before enabling the workflow for real records.

5. FUNCTION IMPLEMENTATION

Deluge is the scripting language used in Zoho Creator and related Zoho services. The following examples are illustrative templates. Replace form link names and field link names with the exact names in the application before using them; confirm the supported workflow event and Deluge syntax in the Creator editor.

5.1 Academic Record Validation — illustrative Deluge

```
// Example: validation logic for an Academic Record form workflow
if(input.Total_Marks <= 0)
{
alert "Total marks must be greater than zero.";
}
else if(input.Passing_Marks < 0 || input.Passing_Marks > input.Total_Marks)
{
alert "Passing marks must be between zero and total marks.";
}
else if(input.Grade_Obtained < 0 || input.Grade_Obtained > 100)
{
alert "Grade obtained must be between 0 and 100.";
}
else
{
// Continue with the configured record submission workflow.
}
```

5.2 Document Verification Follow-up — illustrative Deluge

```
// Example: workflow logic after a document record is updated
if(input.Verification_Result == "Expired")
{
// Update a review flag or notify the assigned reviewer
// using the actual field and notification action configured in the app.
}
else if(input.Verification_Result == "Valid")
{
// Mark the document as ready for the next review step.
}
```

5.3 Admission Decision Workflow — illustrative Deluge

```
// Example: check the admission decision before final processing
if(input.Decision_Outcome == "Accepted" && input.Enrollment_Deadline == null)
{
alert "Enter an enrollment deadline for an accepted application.";
}
else if(input.Decision_Outcome == "Rejected")
{
// Apply the configured rejected status and communication process.
}
else if(input.Decision_Outcome == "Accepted")
{
// Apply the configured accepted status and next-step process.
}
```

Implementation note: The examples demonstrate the intended logic and may require syntax or field-name adjustments in the target Zoho Creator application. Use the platform's form validation or workflow actions supported by the selected event, and test in a development copy before deployment.

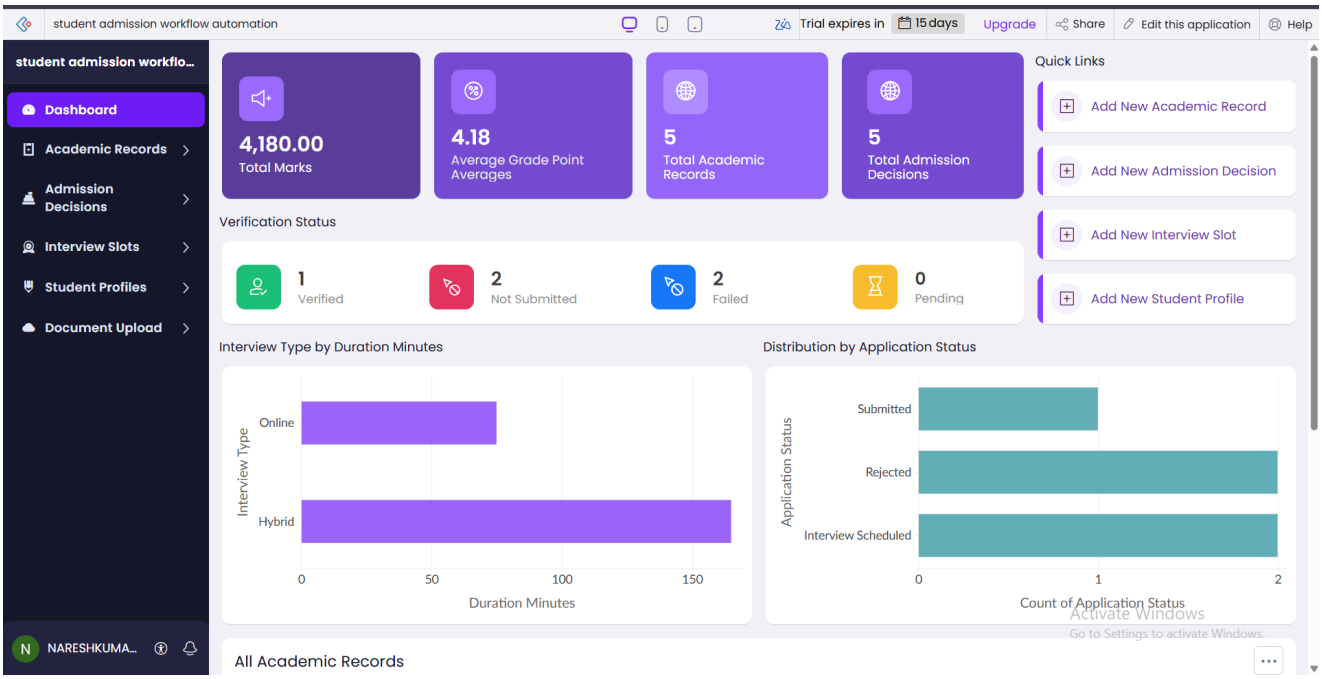
6. RESULT

The proposed Deluge workflow design provides a structured approach to automating the student admission process. It defines workflow triggers and actions for student profiles, academic records, document verification, interview slots, and admission decisions. With correctly configured Deluge functions, the system can reduce repetitive manual checks, improve data consistency, make follow-up tasks easier to track, and keep dashboard metrics aligned with the latest records.

7. APPLICATION SCREENSHOTS

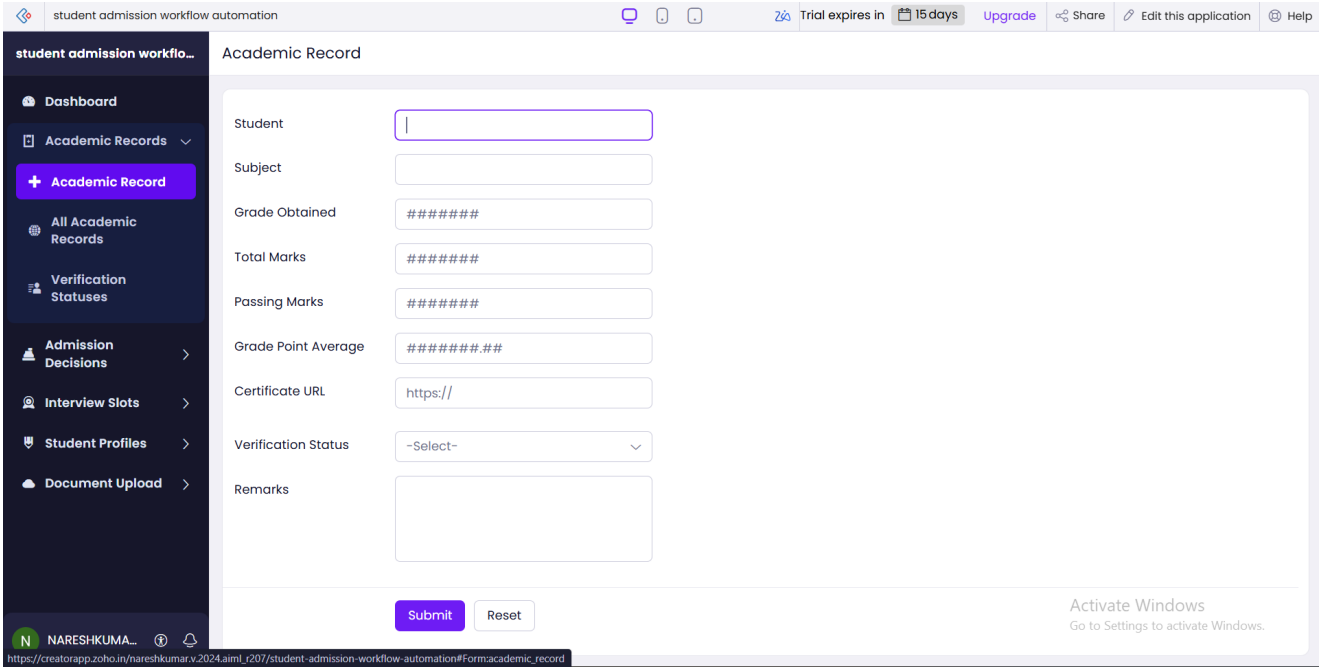
The screenshots below document the dashboard, academic record entry and list, student profile entry and list, and document upload report in the Student Admission Workflow Automation application.

Screenshot 1



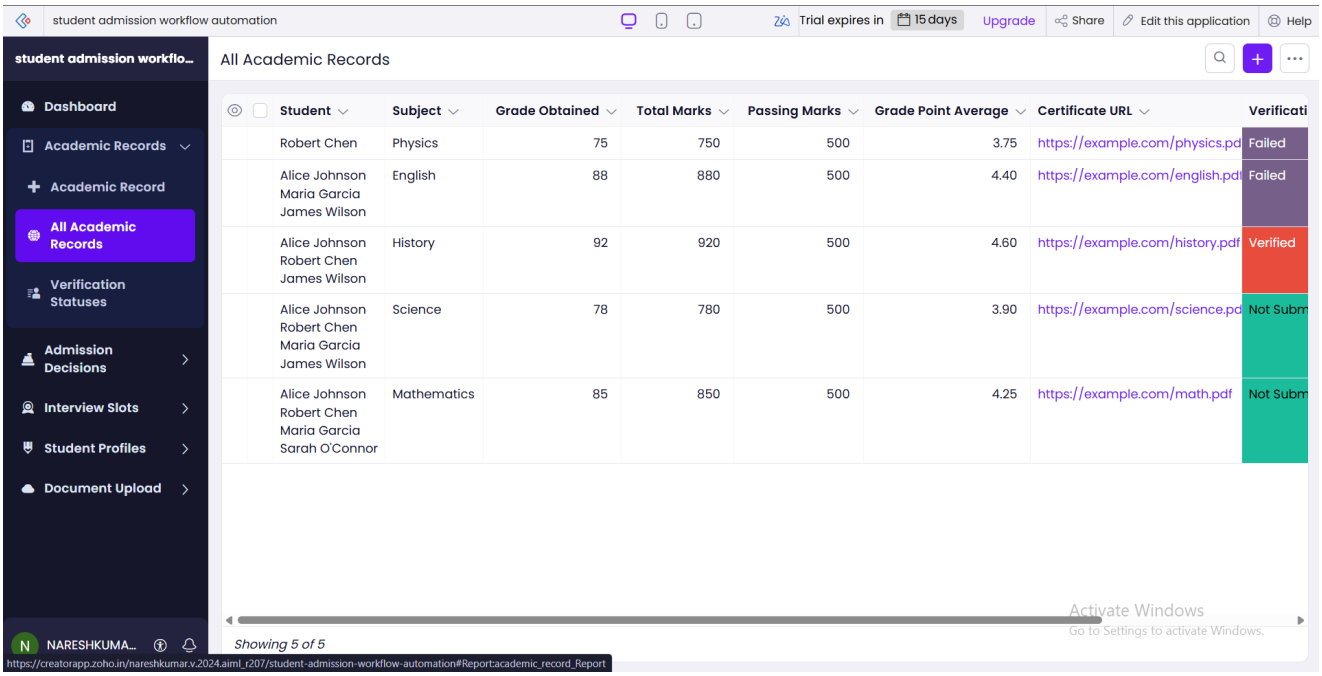
Dashboard with marks, GPA, academic record and admission decision KPIs, verification status, and distribution charts.

Screenshot 2



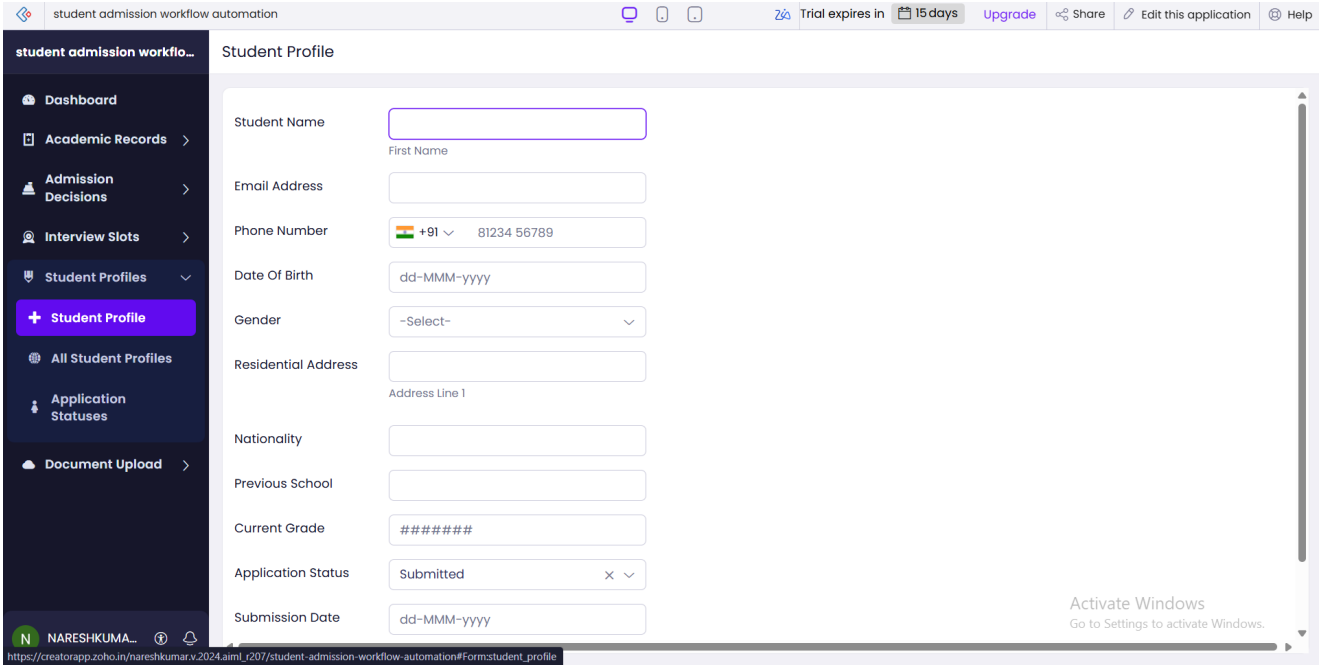
Academic Record form with student, subject, marks, GPA, certificate URL, verification status, and remarks.

Screenshot 3



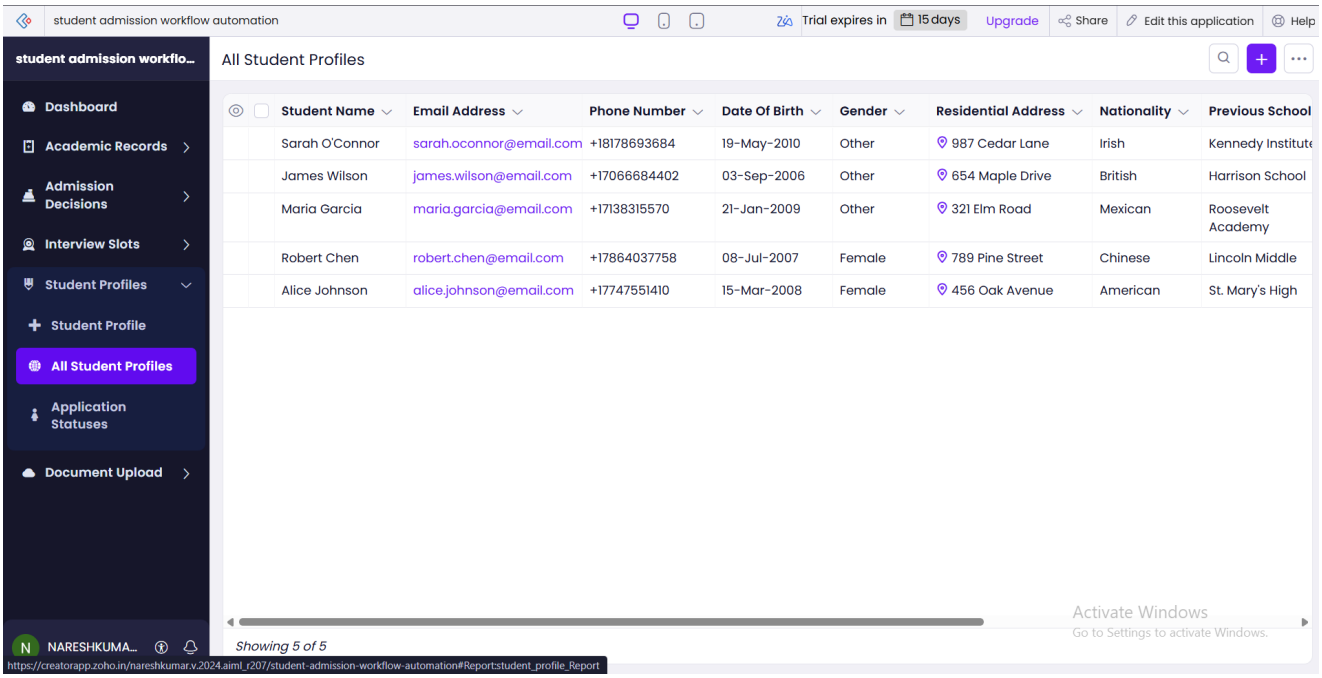
All Academic Records report with grade, marks, GPA, certificate URL, and verification result.

Screenshot 4



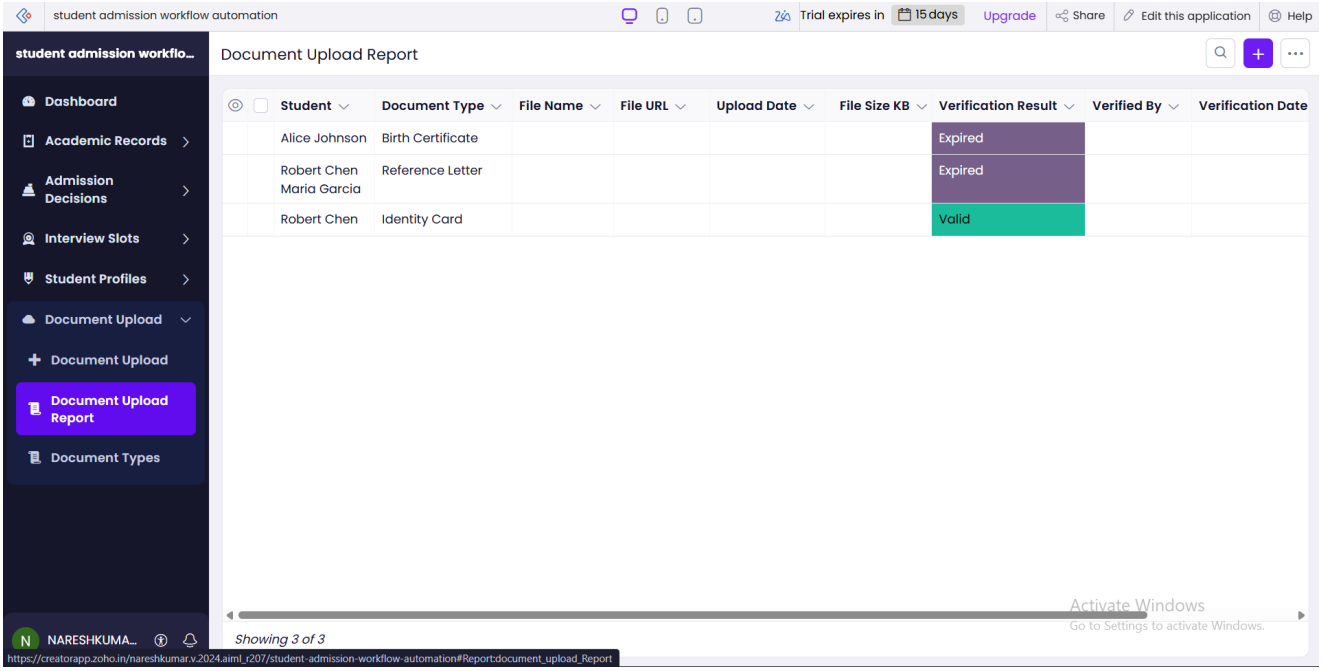
Student Profile form with personal details, previous school, current grade, and application status.

Screenshot 5



All Student Profiles report showing student contact and profile details.

Screenshot 6



Document Upload Report showing document type and verification result.